Technical specification (Requirements specification)

for an intelligent system for monitoring knowledge acquisition

DATERA EDU

1. General information

The System checks which topics a student has learned and which not, and prepares reports for the teacher, the student and the parent. It works like this: the teacher first uploads study materials (text, PDF, presentations) and the System splits them into topics; then the teacher uploads a recording of a one-to-one lesson, and the System checks, topic by topic, whether the student answered correctly, comparing the answers with the material. The MVP, ready by early March 2027, does exactly this: materials, recording, analysis, reports and a knowledge map. Tests on gaps, weekly summaries and notifications are stage 2 (March - April). The MVP runs on ready-made models; fine-tuning of the analysis model is planned for late winter - early spring: first a trial run on open data, then on teachers' corrections.

1.1. System name

- Full name: Development of an intelligent system for monitoring students' knowledge acquisition using speech recognition and large language model technologies for analysing teacher-student dialogue (Russian: Разработка интеллектуальной системы контроля усвоения знаний обучающихся с использованием технологий распознавания речи и больших языковых моделей для анализа диалога преподавателя и ученика).
- Working (commercial) name: DATERA EDU. In this document - the System.

1.2. Grounds for development

Graduation (diploma) thesis. The topic is approved by university order; the order number and date are added after approval.

1.3. Team and roles

Table 1 - Team

Member	**Role**	**Area of responsibility**
David Khamitskiy	Project manager, product manager, ML/AI	Requirements, priorities, acceptance; ML pipeline (audio cleaning, speech recognition, diarization); prompts, quality evaluation and LLM fine-tuning
Tairkhan Yermek	Backend developer	Go API, database, task queue, file storage, authorization, integration with the ML service
Yermakhan Khamitov	Frontend developer	Web interface of the three portals, progress charts, recording player with timestamps, taking tests

Task distribution is reviewed once a month and recorded in YouTrack.

1.4. Timeline

Table 2 - Key dates

Event	Date
Specification approved; development starts	November 2026
Fine-tuning of the analysis model (trial run on open data, then on teachers' corrections) and comparison with a cloud LLM	late winter - early spring 2027
MVP ready: recording upload, analysis, reports (on ready-made models)	early March 2027
Stage 2: tests, summaries, notifications	March - April 2027
Pilot and metrics collection	March - April 2027
Thesis report (explanatory note)	April - May 2027

1.5. Regulatory framework

- GOST 34.602-2020 "Technical specification for the creation of an automated system" - document structure.
- Law of the Republic of Kazakhstan "On Personal Data and Their Protection" - processing of voice recordings, including those of minors.
- Departmental guidelines for formatting the diploma thesis.

2. Terms and abbreviations

Table 3 - Terms and abbreviations

Term	Definition
ASR	Automatic speech recognition: converting audio into text
Whisper	An open speech recognition model; its fast version, faster-whisper, is used
Noise suppression, VAD	Removing noise and echo from audio; finding the parts of a recording where speech is present
Diarization	Labelling a recording by speaker: who speaks and when
Transcript	The text of a lesson with timestamps and speaker labels
LLM	A large language model that analyses the lesson transcript
Fine-tuning	Additional training of a ready-made model on labelled data; LoRA / QLoRA is a method that changes only a small part of the parameters, so one GPU is enough
Mixed speech	Switching from one language to another within a lesson or a phrase
Topic	A unit of material ("Present Perfect", "Quadratic equations") by which learning is assessed
Mastery index	A number from 0 to 100 % showing how well a topic is learned; the calculation method will be refined based on experiments (FR-8)

Term	Definition
Knowledge map	A student's topics in a subject with their index and colour status
WER, DER	The share of wrongly recognised words; the share of recording time with a wrong speaker label
Teacher's materials	Study files for a subject (PDF, documents, presentations, text) that the teacher uploads in advance; the topic list is built from them and the student's answers are checked against them
Anonymisation	Replacing names and other data that could identify a person with labels ("Teacher", "Student 1")
LLM adapter	A layer of the software through which the System calls a language model; the model is changed by configuration, without changing code
MVP, PD	Minimum viable product version of the System; personal data

3. Purpose and goals of the system

The System takes over the manual review of lessons: the teacher and the parent see which topics the student answered correctly and which not, and do not need to rewatch whole recordings. The System's assessment is a suggestion; the final word stays with the teacher (FR-8.7).

3.1. Goals

- Reduce the teacher's time spent reviewing a lesson and preparing for the next one.
- Give the parent a clear picture of the child's progress without special terminology.

3.2. Tasks

- Obtain clean audio from a lesson recording and convert speech to text with timestamps and speaker separation.
- Upload the teacher's materials and split them into topics; check the student's answers on each topic against the material.
- Post-lesson reports for the teacher, the student and the parent.
- The student's knowledge map and its history; in stage 2 - tests on topics not yet learned (questions can be drawn from the teacher's material) and weekly summaries.

3.3. Scientific and practical component of the thesis

- Comparison of noise suppression models by their effect on WER.
- Selection and evaluation of a "recognition + diarization" combination on lesson recordings in English, Russian and mixed English-Russian speech.
- A control WER measurement on several recordings of Kazakh lessons - to justify moving Kazakh to the perspective stage (Section 4.4).
- Method for calculating the mastery index: comparing several variants (three levels or a 0-100 % scale, different weights) by agreement with the teacher's assessment (NFR-5.5).

– Evaluation of the quality of LLM analysis and of how recognition errors affect it: the LLM analyses manually verified and recognised text of the same recordings, and the assessments are compared.
– Fine-tuning of the analysis model: a trial run on open datasets (late winter - early spring), then on teachers' corrections (stage 2), and comparison with the base model and a cloud LLM (Section 5.5).

4. Description of the automation object and users

The object of automation is knowledge control in online and in-person lessons. The System has three portals: organization, student and parent. Clients: online schools and learning centres, individual tutors, specialised schools.

The list of subjects is not limited: topics are taken from the materials the teacher uploads, so the System is not tied to a subject. The pilot tests 2-3 subjects (Section 10.1). The key MVP scenario is English lessons, including explanations in Russian, and lessons in Russian. The choice of languages is driven by recognition accuracy (Section 4.4); how much demand there is for English in private tutoring in Kazakhstan will be checked in the pilot.

4.1. Current process and its problems

– The teacher recalls by hand what was covered in previous lessons; when the teacher changes, the history is lost.
– The parent learns about progress from the teacher and from homework; there is no complete picture by topic.

4.2. User roles

Table 4 - User roles

Role	Portal	Who they are	Main actions
Organization administrator	Organization	Head of a centre or school	Manages teachers, students and the question bank; sees analytics for all students
Teacher	Organization	Centre staff member or tutor	Uploads materials and confirms topics, uploads recordings, reads reports, changes the System's assessment if they disagree; in stage 2 - assigns and edits tests
Student	Learner	Learner	Views reports, the knowledge map and recordings; takes tests (stage 2)
Parent	Parent	Legal guardian	Views reports and the knowledge map of their children, gives consent to processing of recordings
System administrator	Internal	Development team	Manages organizations, models and prompts, monitors the queue

4.3. Lesson format

- Individual (one-to-one) - the main scenario, fully included in the MVP.
- Group (several students on one recording) - perspective: in the MVP the System works with individual lessons. Matching voices to students and recognising students by voice are described in Section 11 (FR-6.5, FR-6.8).

4.4. Lesson languages

Table 5 - Lesson languages by stage

Language	**Status**	**Notes**
English	MVP, priority 1	The main language; open models give their best recognition accuracy for it
Russian	MVP, priority 2	Most lessons in Kazakhstan explain the material in Russian
Mixed English-Russian speech	MVP	Tasks and answers in English, explanations in Russian. The same multilingual model; quality is measured separately (NFR-5.1)
Kazakh and mixed Kazakh-Russian speech	Perspective	Left out of the MVP because of low recognition accuracy (see below)

Why Kazakh is moved to the perspective stage. The literature review done within the project showed that Kazakh speech is recognised noticeably worse than Russian and English, even in favourable conditions:

- **Z. Makhataeva, N. Atymtay, R. Meiramov, G. Nauryzbaikyzy, K. Sadirova, H. A. Varol [1]:** Whisper fine-tuned by ISSAI on Kazakh reduced WER on the test set from 21.55 to 8.84 %. But when 38 people read prepared texts aloud, WER was 26.32 % for Kazakh, 11.86 % for Russian and 9.65 % for English; the difference is significant. On live lessons with noise, errors are likely to be higher.
- **S. Mussakhojayeva, K. Dauletbek, R. Yeshpanov, H. A. Varol [2]:** the best result for Kazakh is 5.7 % WER on one test corpus (KSC) and 28.6 % on another (Common Voice), with different models. Published figures depend heavily on the dataset and transfer poorly to new recordings.
- The review found no work that evaluated diarization and recognition of Kazakh or mixed Kazakh-Russian lessons [3]. The quality of the whole chain for Kazakh is unknown in advance, and the team cannot check it before March 2027.

A recognition error passes into the analysis: a wrongly recognised word of the student may be judged by the LLM as a wrong answer. So the MVP includes the languages with more reliable recognition, and Kazakh is added after the MVP (Section 11).

5. System-wide requirements

The System has three parts: a web client, a Go API and a Python ML service, which exchange tasks through a queue. A lesson recording passes through the processing stages in sequence *(Figure 1)*, each with its own model or program.

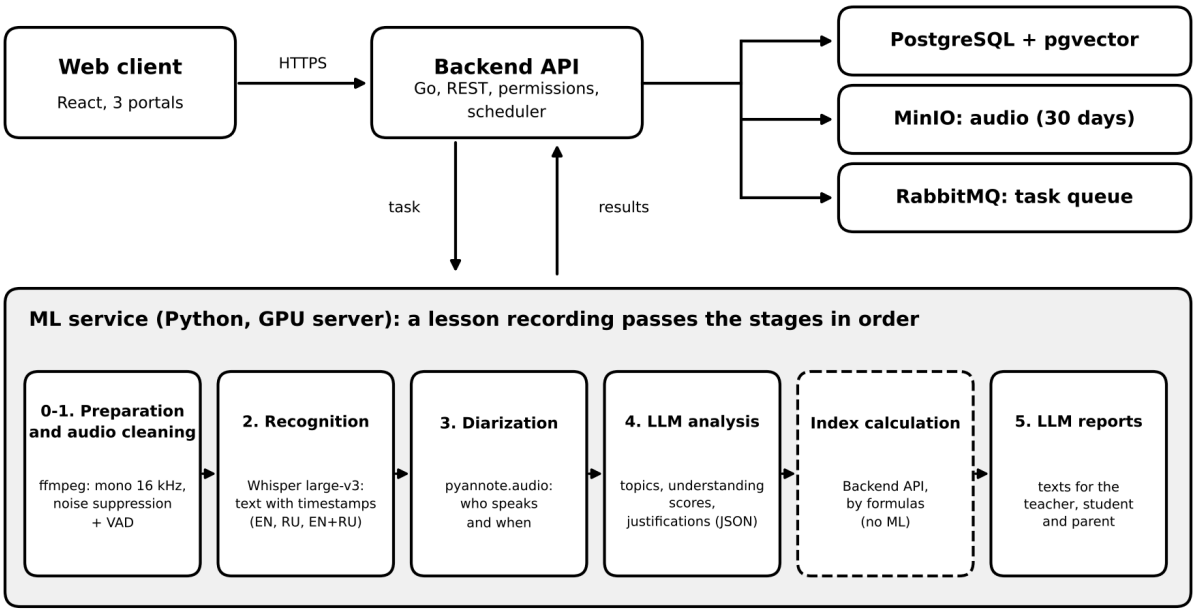


Figure 1 - System architecture and recording processing pipeline

5.1. System components

Table 6 - Subsystems

Subsystem	**Purpose**
Web client	Portals for the organization, the student and the parent
Backend API (Go)	Accounts and permissions, learning structure, file intake, index calculation, reports, scheduler (audio deletion); in stage 2 - tests, summaries, notifications
ML service (Python)	Audio cleaning, recognition, diarization, calls to the LLM for analysis and report writing
Storage	PostgreSQL - data and topic vectors; MinIO - audio; RabbitMQ - task queue; Redis - sessions

5.2. Recording processing pipeline

Table 7 - Recording processing stages

#	Stage	What it does	Model / tool	Result
0	Preparation	Extracts audio from video, converts to mono 16 kHz, normalises loudness	ffmpeg	Audio file
1	Audio cleaning	Removes noise and echo, marks the parts with speech	Noise suppression (DeepFilterNet / Demucs / RNNoise - chosen after comparison) + VAD	Cleaned audio
2	Recognition	Converts speech to text with word timestamps; understands English, Russian and their mix	Whisper large-v3 (faster-whisper)	Text with timestamps
3	Diarization	Splits the recording by speaker, merges it with the text, identifies the teacher	pyannote.audio	Transcript "speaker - time - text"
4	LLM analysis	Finds the places in the lesson where topics from the material were discussed; checks the student's answers against the material; assigns a status to each topic with a justification and timestamps; writes a summary	LLM via adapter (Section 8.2); in stage 2 - the fine-tuned model	Result in JSON
-	Index calculation	Calculates the index and topic statuses by formulas, finds "stuck" topics and risk, creates the task for reports	Backend API (no ML)	Index, knowledge map
5	LLM reports	From the analysis and the index writes three texts: for the teacher, the student and the parent	The same LLM	Report texts
-	Test and notifications (stage 2)	Builds a test on topics with a low index, creates notifications	Backend API (no ML)	Test, notifications

Stages 0-4 are run by the ML service, which returns the result to the API; the API calculates the index and gives the ML service the task of writing reports (stage 5). The numbers (index, statuses, risk) are calculated only by code using formulas; in reports the LLM does not change them but explains them in plain language.

5.3. Design decisions

– **Go + Python.** Business logic is in Go: the team's backend developer writes in it, and it suits a service with many requests. ML is a separate Python service, because

recognition, diarization and noise suppression models are mostly available in the Python ecosystem.

- **Task queue.** An upload does not wait for the analysis to finish; a failure of the ML service does not stop the portals.
- **One multilingual recognition model.** Whisper large-v3 understands English, Russian and their mix without switching models.
- **Topics come from the teacher's materials.** The LLM proposes a list of topics from the uploaded files and the teacher confirms it. This way the System is not tied to a subject, and the "right answer" is defined by the material, not by the model's general knowledge.
- **LLM adapter and anonymisation.** The model is changed by configuration; only text without names goes to an external service.
- **Rules on top of the LLM.** The index, statuses and risk are calculated by code, not by the LLM, so results are reproducible. Formulas and thresholds are configurable without code changes and are checked on a test set before the pilot.

5.4. Operating modes

The portals work around the clock. If the ML service or the LLM is unavailable, recordings wait in the queue and are processed after recovery.

5.5. Ready-made models in the MVP and fine-tuning of the analysis model

The System is built on ready-made models, and the analysis model (stage 4) is additionally fine-tuned, because the main result - the assessment of topic understanding - depends on it. The work goes in three steps, so that the first fine-tuning result appears before the MVP is ready and does not depend on the pilot schedule.

- **Step 1 - MVP on ready-made models (November 2026 - early March 2027).** Noise suppression, Whisper, pyannote.audio and a cloud LLM are used without training. The "Confirm" and "Change" buttons in the report (FR-8.7) are part of the MVP, so teachers' assessments accumulate from the very first lesson.
- **Step 2 - trial fine-tuning on open data (late winter 2027).** An open LLM is fine-tuned on open English-language datasets of teacher-student dialogues: MathDial [4] (about 3 thousand dialogues of real teachers with a language model playing a student who makes typical mistakes) and NCTE Transcripts [5] (1,660 transcripts of maths lessons in grades 4-5 with utterance labels). The aim is to test the process and get a first comparison with the base model. These datasets describe teacher actions, not checking answers against material, so what is tested here is the process, not the final quality of the assessment; the comparison task (for example, detecting the type of a teacher's utterance) is fixed before training starts. The licences of the datasets are checked in advance.
- **Step 3 - fine-tuning on project data (early spring 2027, as examples accumulate).** The model from step 2 is fine-tuned on examples from the pilot:
- Data: an example is a lesson fragment, a piece of material and the teacher's assessment (confirmed or changed by the teacher in the report, FR-8.7). No separate labelling is

needed. The goal is at least 300 examples, preferably 500; if there is less data, training uses what is available, and the thesis reports the result with a caveat.

- Control set: 5-10 lessons where the teacher separately marked whether the student answered correctly on each topic (FR-16.2); it is used to measure the System's accuracy (NFR-5.5).
- Split by lessons, not by fragments: lessons of the control set do not go into training.
- Model: an open LLM with 7-9 billion parameters (Qwen or Llama family), fine-tuned with LoRA / QLoRA [6, 7] on one GPU.
- Comparison of the base open model, the fine-tuned model and the cloud model on the control set: agreement with the teacher (NFR-5.5), share of valid JSON (NFR-5.6), speed, cost.
- Deployment: if the fine-tuned model falls behind the cloud one on NFR-5.5 by no more than 5 percentage points, it is connected through the same adapter, and transcripts do not leave the System's server. If it falls behind by more, the cloud model stays, and the comparison is described in the thesis.

Fine-tuning of the noise suppression and recognition models is outside the scope of the thesis (Section 11).

6. Functional requirements

The requirements are grouped into modules FR-1 - FR-16; the codes have not changed. There are three priorities. "MVP" - mandatory by March 2027: the teacher uploads a recording of a one-to-one lesson, and the System analyses it and shows the result. "Stage 2" - March - April 2027, within the scope of the thesis: tests on gaps, summaries, notifications and model fine-tuning; fine-tuning runs from late winter - early spring (table at the end of the section). "Perspective" - outside the scope of the work, listed in Section 11. The MVP is designed for individual lessons in English, Russian and mixed English-Russian speech.

FR-1. Registration, login and access control

Code	Requirement
FR-1.1	Registration of an organization by email and password; a tutor registers as an organization of one person
FR-1.2	The organization administrator creates accounts for teachers and students; access is given by an invitation link
FR-1.3	A parent is linked to one or more students by an invitation link
FR-1.4	A parent sees only their own children, a teacher - their own students, the organization administrator - all students of the organization

FR-2. Consent to processing of recordings

Code	Requirement
FR-2.1	Before the first recording upload the System requests electronic consent: from the parent if the student is under 18, otherwise from the student. Without consent, recordings of this student cannot be uploaded

Code	Requirement
FR-2.2	The System stores who gave consent, when, and for which version of the text
FR-2.3	Separate consent to the use of an anonymised transcript for fine-tuning (FR-16); without it the lesson does not enter the training set
FR-2.4	Withdrawal of consent: the student's recordings, transcripts and reports are deleted within 3 working days; in the MVP the system administrator does this on request

FR-3. Learning structure

Code	Requirement
FR-3.1	Creating subjects with any name
FR-3.2	A "student - teacher - subject" link with the language of instruction (English or Russian)
FR-3.4	Uploading materials for a subject: PDF, DOCX, PPTX, TXT, MD, up to 20 MB per file; scans without a text layer are not supported in the MVP
FR-3.5	Text is extracted from the materials; the LLM proposes a list of topics for the subject (name and short description)
FR-3.6	The teacher confirms, renames, merges or deletes topics and can add their own; only confirmed topics take part in the analysis
FR-3.7	Materials are visible only to their own organization and are stored on the System's server; a file can be replaced, after which the topic list is updated with the teacher's confirmation

FR-4. Lesson recording upload

Code	Requirement
FR-4.1	MP4, MOV, MKV, WEBM, MP3, WAV, M4A, OGG files up to 2 GB and 90 minutes
FR-4.2	On upload the student, subject, date and lesson language are specified: English, Russian or "mixed / detect automatically" (default)
FR-4.3	Upload progress is shown
FR-4.4	Audio is extracted from the video, and the video file is deleted immediately
FR-4.5	Processing status: "Uploaded" → "Audio cleaning" → "Recognition" → "Speaker separation" → "Analysis" → "Reports" → "Done" or "Error"
FR-4.6	Processing can be restarted after an error
FR-4.7	Audio is deleted 30 days after upload; the transcript and reports are kept

FR-5. Audio preparation and cleaning (stages 0-1)

Code	Requirement
FR-5.1	Converting audio to mono 16 kHz and loudness normalisation
FR-5.2	Noise suppression by an ML model; the model is chosen by comparing candidates (DeepFilterNet, Demucs, RNNoise) by WER reduction

Code	Requirement
FR-5.3	Speech detection (VAD): long silence is skipped to reduce the risk of "made-up" text on pauses
FR-5.4	Noise suppression can be switched off for a recording if it worsens recognition
FR-5.5	Cleaned audio is stored for the player within the period set in FR-4.7

FR-6. Speech recognition and speaker separation (stages 2-3)

Code	Requirement
FR-6.1	Recognition of English, Russian and mixed English-Russian speech with phrase and word timestamps (Whisper large-v3)
FR-6.2	Diarization (pyannote.audio): segments with a speaker label
FR-6.3	Merging recognition and diarization into a "speaker - time - text" transcript
FR-6.4	Automatic identification of the teacher; the teacher can correct the labelling, after which the analysis is run again
FR-6.6	Recording the student's pauses before answering (by default longer than 5 seconds, the threshold is configurable), requests to repeat and repetitions; they are passed to the analysis
FR-6.7	Viewing the transcript with highlighted roles

FR-7. AI analysis of the lesson (stage 4)

Code	Requirement
FR-7.1	The LLM finds in the transcript the places where confirmed topics of the subject were discussed (FR-3.6) and marks them with start and end timestamps
FR-7.2	For each topic the LLM checks the student's answers against the material (FR-3.4) and assigns a status; in the MVP there are three levels: "not learned", "partly learned", "learned" (D = 0, 50 or 100 %). Justification: correct answers, mistakes, requests to repeat, refusals to answer - each with a timestamp and a reference to the place in the material. The form of assessment will be refined based on experiments
FR-7.8	If something was discussed in the lesson that is not in the material, the fragment is marked "outside the material", is not included in the index, and is shown to the teacher with a suggestion to add a topic
FR-7.3	Counting the teacher's repeated explanations of one topic
FR-7.4	A short lesson summary and recommendations for the next lesson
FR-7.5	The LLM response arrives as JSON in a fixed schema; on error the request is repeated up to 3 times, then the recording gets the status "Error"
FR-7.6	Before any text is sent to an external LLM (stages 4 and 5), names and other data that could identify a person are replaced with labels ("Teacher", "Student 1"); the real names are put back into the finished texts by the API
FR-7.7	Work with the LLM goes through an adapter: the provider or model (cloud, local, fine-tuned) is changed by configuration, without changing code

FR-8. Mastery index and knowledge map

The topic mastery index combines the assessment of answers in the lesson (D) and the share of correct answers in tests on the topic (T):

$$I = 0.4 \cdot D + 0.6 \cdot T$$

If there was no test on the topic, I = D. Tests appear in stage 2, so in the MVP the index equals D. The formula, weights, thresholds and the way D is obtained are an initial version: they are checked on a test set, compared with the teacher's assessment and refined based on experiments. In the MVP D takes the values 0, 50 or 100 % (FR-7.2).

Table 8 - Topic colour statuses

Colour	**Index**	**Status**
Red	below 50 %	Not learned - problem
Yellow	50-79 %	Learned, needs reinforcement
Green	80 % and above	Consolidated

Code	**Requirement**
FR-8.1	Index calculation after each lesson
FR-8.2	Knowledge map for a subject: topics with colour, index, date of the last review and direction of change
FR-8.6	The knowledge map history is stored per student and is not lost when the teacher changes
FR-8.7	The teacher can confirm a topic status or change it (comment optional); the index and map are recalculated, and the change goes into the history; the final word always belongs to the teacher

FR-9. Reports (stage 5)

Code	**Requirement**
FR-9.1	After the analysis three kinds of lesson report are produced: for the teacher (detailed, with recommendations), for the student (what to review, in plain language), for the parent (short, without terms). Reports for the student and the parent are written in a supportive tone and without judgmental labels such as "weak student". Each kind is written in the recipient's language (FR-9.2), so with different languages there may be more than three texts
FR-9.2	The user chooses the report language (Russian or English) in their profile, independently of the lesson language
FR-9.3	Teacher's report: summary, topics with index and status, moments of difficulty, number of repeated explanations, recommendations
FR-9.4	A report item with a timestamp opens the player at that moment; after the audio is deleted, a transcript fragment is shown

FR-11 - FR-13. Teacher, student and parent portals

Code	Requirement
FR-11.1	Teacher: list of students with a progress indicator - average index, number of red topics, date of the last lesson
FR-11.5	Teacher: a student's lesson history - recordings, transcripts, reports
FR-12.1	Student: feed of lessons with reports and the recording player
FR-12.2	Student: knowledge map and a "What to review" list in plain language
FR-13.1	Parent: switching between children if there are several
FR-13.2	Parent: lesson reports without professional terms
FR-13.3	Parent: the child's knowledge map by topic

FR-15. System administration

Code	Requirement
FR-15.2	Monitoring of the queue and errors at each stage
FR-15.3	Prompt versions and choice of LLM; a report stores which model and prompt version produced it

FR-16. Data preparation and fine-tuning of the analysis model (part of the MVP)

Code	Requirement
FR-16.0	Trial fine-tuning of an open LLM on open dialogue datasets with a report comparing it to the base model (Section 5.5, step 2)
FR-16.1	Each confirmation or change of a status (FR-8.7) is saved as an example: fragment, material, the System's assessment and the teacher's assessment; only anonymised data and only lessons with consent under FR-2.3
FR-16.2	Control set: 5-10 lessons where the teacher independently marked the correctness of answers on each topic; these lessons do not go into training

Stage 2 requirements (March - April 2027)

This is what is done after the MVP while the pilot runs. If the schedule slips, this list is cut, not the MVP (Section 9.1).

Code	Requirement
FR-1.5	Password recovery by email
FR-3.3	Transferring a student to another teacher with the full history preserved
FR-8.3	Chart of the topic index and of the average subject index over time
FR-8.4	"Stuck" topic: index below 50 % in three consecutive lessons where the topic was covered
FR-8.5	Risk level for a topic (low / medium / high) by rules based on the index and its dynamics
FR-9.5	A weekly summary every Monday for the parent and the teacher: lessons, learned and problem topics, recommendations

Code	Requirement
FR-9.7	The organization chooses: reports are published to the student and the parent immediately (default) or after the teacher has reviewed them
FR-10.1	The organization's question bank: subject, topic, language, difficulty (1-3), type, options, correct answer, explanation
FR-10.2	The bank is filled manually, by import from CSV or Excel using a template, or by LLM generation from the teacher's materials; the teacher reviews generated questions before use
FR-10.3	Bank questions are linked to the subject's topics (FR-3.6), so a topic from the lesson analysis immediately finds its questions
FR-10.4	After a lesson a test of 5-15 questions is built on topics with an index below 80 %; the lower the index, the more questions; questions are in the student's language of instruction
FR-10.5	If there are not enough questions, the teacher gets a notification
FR-10.6	The teacher can view, change or cancel the test before it is issued; if the teacher does nothing, the test is issued automatically
FR-10.7	The student takes the test in the browser before the next lesson; checking is automatic; after submission a review is shown
FR-10.8	Test results recalculate the topic indexes (FR-8.1)
FR-11.2	Teacher: a "Needs attention" block - stuck topics, high risk, tests not taken
FR-12.3	Student: active tests, history and results
FR-13.4	Parent: access to lesson recordings if the organization's settings allow it
FR-14.1	Notifications in the portal: report ready, test assigned, test submitted, weekly summary ready, topic became "stuck"
FR-15.4	Access log for recordings and transcripts: who opened them and when
FR-16.3	Splitting examples into training and control sets by lesson
FR-16.4	Fine-tuning an open LLM (7-9 billion parameters) with LoRA / QLoRA; saving the model version and parameters
FR-16.5	Comparing the base, fine-tuned and cloud models on the control set by NFR-5.5 and NFR-5.6, with a report
FR-16.6	Connecting the fine-tuned model through the adapter (FR-7.7) when the condition in Section 5.5 is met

7. Non-functional requirements

In short: the report for a one-hour lesson is ready in 30 minutes, the pilot load is up to 10 lessons a day, and student data is stored on servers in Kazakhstan.

NFR-1. Performance

Code	Requirement
NFR-1.1	The report for a 60-minute recording is ready no later than 30 minutes after upload, if there are no other recordings in the queue

Code	Requirement
NFR-1.2	At least 10 recordings per day; when uploaded at the same time, recordings are processed in turn
NFR-1.3	API response to typical requests - no more than 500 ms for 95 % of requests
NFR-1.4	A portal page loads in no more than 3 seconds; up to 30 users at the same time

NFR-2. Reliability

Code	Requirement
NFR-2.1	Tasks are kept in a durable queue and are not lost when services restart
NFR-2.2	Processing continues from the last completed stage, not from the beginning
NFR-2.3	Each stage is retried up to 3 times on failure, then the recording gets the status "Error" with the reason
NFR-2.4	Database backup - daily, kept for 7 days
NFR-2.5	Availability during the pilot - at least 95 % per month

NFR-3. Security

Code	Requirement
NFR-3.1	Data is exchanged only over HTTPS (TLS 1.2 or higher)
NFR-3.2	Passwords are stored only as a hash (bcrypt or Argon2)
NFR-3.3	JWT: the access token lives 15 minutes, there is a refresh token
NFR-3.4	Permissions are checked on the server with every request, not only in the interface
NFR-3.5	Recordings are not available by a direct link; playback and download use a signed link valid for 15 minutes
NFR-3.6	Audio files are stored encrypted
NFR-3.7	Login is blocked for 15 minutes after 5 failed attempts
NFR-3.8	Protection against the vulnerabilities in the OWASP Top 10 (injection, XSS, CSRF)

NFR-4. Personal data

Code	Requirement
NFR-4.1	The database and files with personal data are hosted on servers in the Republic of Kazakhstan
NFR-4.2	Only anonymised text goes to external services (cloud LLM) (FR-7.6); audio is not transferred outside the System's server
NFR-4.3	Storage: audio - 30 days; transcripts, reports and the knowledge map - until the student's account is deleted or consent is withdrawn
NFR-4.4	Recordings of minors are processed only with the consent of a legal guardian (FR-2.1)

Code	Requirement
NFR-4.5	The training set consists only of anonymised texts with consent under FR-2.3; it is stored on the System's server. For fine-tuning, a university server or a rented GPU in Kazakhstan is allowed; after training the data is deleted from it
NFR-4.6	The consent text and the privacy policy are legally reviewed before the pilot starts

NFR-5. Quality of ML components

Target values are checked on a test set of at least 20 real recordings (at least 10 in English, of which at least 5 with mixed English-Russian speech, and at least 10 in Russian) with a reference text and speaker labels, and for 5-10 recordings also with the teacher's assessment of topics (FR-16.2). The values for mixed speech and DER are preliminary and will be refined after the first measurement.

Code	Metric	Target value
NFR-5.1	WER: English / Russian / mixed speech	no more than 15 % / 20 % / 25 %
NFR-5.2	Effect of noise suppression	WER after cleaning is lower than without it on noisy recordings
NFR-5.3	DER on individual lessons	no more than 20 % (preliminary)
NFR-5.4	Correct identification of the teacher	at least 95 % of recordings
NFR-5.5	Agreement of the topic status (red / yellow / green) with the teacher's assessment on the control set	at least 70 % of topics
NFR-5.6	LLM responses that pass JSON schema validation on the first attempt	at least 95 %
NFR-5.7	Fine-tuned model (stage 2): agreement with the teacher on the control set	higher than the same model without fine-tuning
NFR-5.8	Share of statuses that the teacher left unchanged in the pilot	at least 70 %

NFR-6. Compatibility and usability

Code	Requirement
NFR-6.1	The two latest versions of Google Chrome; Edge, Firefox and Safari desirable
NFR-6.2	Responsive layout: screens from 360 px
NFR-6.3	The interface is in Russian; reports in Russian or English (FR-9.2); texts are kept in translation files
NFR-6.4	The parent's report has no professional terms and is understandable without training
NFR-6.5	Uploading a recording and opening a report take no more than 3 steps from the home page

NFR-7. Maintainability and scalability

Code	Requirement
NFR-7.1	All services run in Docker containers, started with one command (Docker Compose)
NFR-7.2	The API is described in OpenAPI
NFR-7.3	Code is in Git; linters and tests run automatically on every change
NFR-7.4	Unit tests cover at least 50 % of the backend business logic
NFR-7.5	Structured logs with a single lesson recording identifier
NFR-7.6	The API keeps no state between requests; ML workers scale by adding instances
NFR-7.7	Experiments are reproducible: versions of models, prompts, data and training parameters are recorded

8. Requirements for types of support

8.1. Software

The choice of each technology is justified in the Technology Comparison.

Table 9 - Technologies

Component	Technology	Purpose
Frontend	React + TypeScript, Vite; shadcn/ui (Tailwind CSS); Recharts; TanStack Query; wavesurfer.js	Three portals, knowledge map, charts, player with jump-to-timestamp
Backend API	Go (Gin or chi - chosen at start), REST + OpenAPI	Users, permissions, lessons, index, reports, scheduler (in stage 2 - tests)
Database	PostgreSQL 16 + pgvector	Data; vectors for topic matching (FR-7.1, in stage 2 - FR-10.3)
Queue, cache	RabbitMQ; Redis	Tasks between the API and the ML service; sessions and the login attempt limit
Files	MinIO (S3-compatible)	Audio with automatic deletion after 30 days
Mail	SMTP service	Password recovery emails (stage 2, FR-1.5)
ML service	Python 3.11, PyTorch (CUDA), ffmpeg	Pipeline stages 0-5
Models	DeepFilterNet / Demucs / RNNoise (chosen after comparison); faster-whisper large-v3; pyannote.audio	FR-5.2, FR-6.1, FR-6.2
LLM and fine-tuning	MVP - cloud LLM through the adapter; stage 2 - an open model via vLLM; Hugging Face Transformers, PEFT (LoRA / QLoRA)	FR-7, FR-9, FR-16

Component	Technology	Purpose
Deployment	Docker, Docker Compose, Nginx (HTTPS)	NFR-7.1

8.2. Choice of LLM

The MVP uses a top-tier cloud LLM (Claude Sonnet or GPT class) with transcript anonymisation. In stage 2 an open Qwen or Llama model fine-tuned on project data is connected through the same adapter.

Table 10 - Cloud and local LLM

Criterion	Cloud LLM (MVP)	Local open LLM (stage 2)
Quality of long-dialogue analysis	High without fine-tuning	Lower without fine-tuning; fine-tuning should close the gap
Personal data	Anonymisation needed (FR-7.6)	Data does not leave the server
Cost at 10 lessons a day	Pay per token, about tens of dollars a month (estimate to be refined against the price list)	Hardware or rental
Speed of adoption	Immediate	Needs data, training, server setup

8.3. System server

The requirements apply to the server that processes recordings; users do not need such hardware.

Table 11 - Server requirements

Resource	Minimum requirements
GPU for running the System	NVIDIA with 16 GB of video memory or more (Whisper, pyannote, noise suppression); 24 GB or more if a local LLM is run
GPU for fine-tuning (stage 2)	24 GB of video memory or more for QLoRA of a 7-9 billion parameter model; a university server or a GPU rented from a provider in Kazakhstan is allowed
CPU / RAM / disk	8 cores, 32 GB, SSD of 500 GB or more
OS and hosting	Ubuntu 22.04 or 24.04 LTS, NVIDIA drivers + CUDA; a server or cloud in Kazakhstan (NFR-4.1); for development a university or team server is allowed

9. Scope and content of work

The MVP is developed from November 2026 to March 2027. Fine-tuning of the analysis model runs in late winter - early spring, without fixed dates: it must not get in the way of the MVP. From March to April the pilot and stage 2 (tests, summaries, notifications) run, and from April to May the thesis is written *(Figure 2)*.

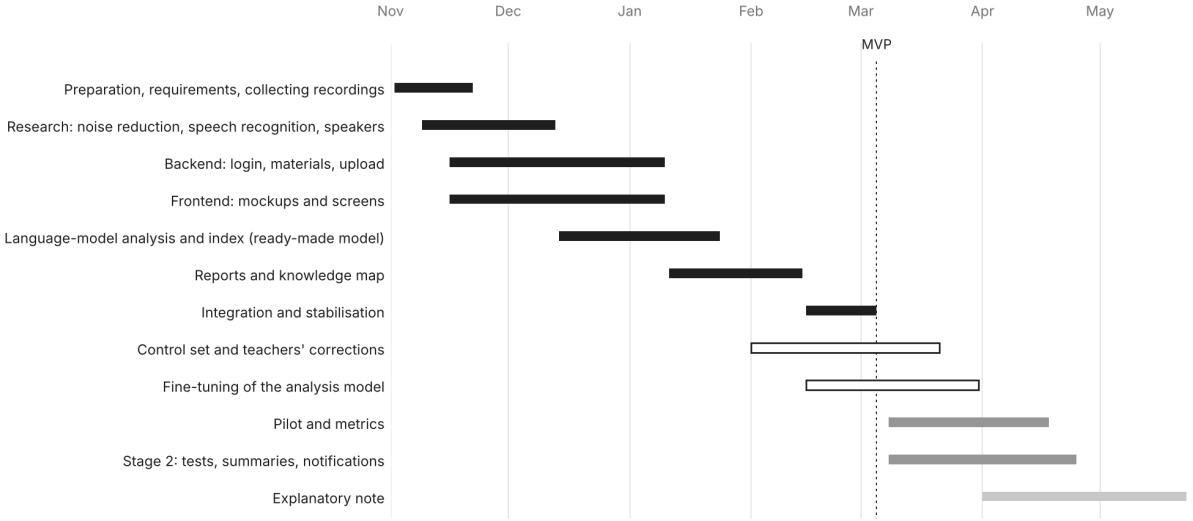


Figure 2 - Work schedule

Table 12 - Work stages

Stage	**Result**	**Responsible**
Preparation, specification, collecting recordings	Approved specification; at least 20 test recordings in English and Russian with consent; repository, Docker, a server with a GPU	Whole team
Model research	Comparison of noise suppression by WER; chosen recognition model; WER and DER on the test set; WER measurement on Kazakh recordings	David Khamitskiy
Backend: login, materials, upload	Registration, roles, learning structure, materials and topics, consents, recording upload, queue, storage (FR-1 - FR-4)	Tairkhan Yermek
Frontend: mockups and screens	Figma mockups; login, student list, recording upload, statuses	Yermakhan Khamitov
LLM analysis and index	Cloud LLM, splitting materials into topics, checking answers, prompts, JSON schema, index and statuses (FR-3.5, FR-7, FR-8); comparison of analysis on verified and recognised text	David Khamitskiy, Tairkhan Yermek
Trial fine-tuning	A model fine-tuned on MathDial / NCTE; comparison with the base model (FR-16.0); late winter - early spring	David Khamitskiy

Stage	Result	Responsible
Reports and knowledge map	Three kinds of reports, knowledge map, player with timestamps (FR-8, FR-9, FR-11 - FR-13)	Tairkhan Yermek, Yermakhan Khamitov
Integration and stabilisation	The end-to-end scenario works; MVP ready (March)	Whole team
Control set and corrections	Assessment of 5-10 lessons by a teacher; accumulation of confirmations and corrections for fine-tuning (FR-8.7, FR-16.1, FR-16.2); at least 300 examples (goal - 500)	David Khamitskiy, pilot teachers
Pilot and metrics	2-3 teachers working with the System; metrics NFR-1 and NFR-5; feedback	Whole team
Stage 2: tests, summaries, notifications	Question bank and tests, weekly summaries, notifications, index chart, risk (FR-8.3 - FR-8.5, FR-9.5 - FR-9.7, FR-10, FR-14)	Tairkhan Yermek, Yermakhan Khamitov
LLM fine-tuning on teachers' corrections	A fine-tuned model; comparison of three models; decision on connecting it (FR-16.4 - FR-16.6)	David Khamitskiy
Thesis	Thesis text and a demonstration stand	Whole team

9.1. Plan risks

- **Recognition on real lessons is worse than expected, especially on mixed speech.** Measure WER on the first recordings in November; if needed compare large-v3 and the turbo variant and mark lessons with lower accuracy.
- **Not enough data for fine-tuning.** Trial fine-tuning on open datasets before the MVP; examples accumulate from teachers' confirmations and corrections from the first lesson. According to the QLoRA authors [6], a small set of high-quality examples may be enough - the project tests this.
- **The fine-tuned model does not catch up with the cloud one.** The cloud model stays in use, and the comparison is described in the thesis as a scientific result.
- **No server with a GPU.** Research is done in Colab or Kaggle, but only on open data and on recordings whose processing abroad all participants have allowed; real lesson recordings stay only on a server in Kazakhstan (NFR-4.1). Fine-tuning is done on a university server or a GPU in Kazakhstan; integration needs a permanent server.
- **Overload of the project manager: he runs the ML pipeline, the LLM, fine-tuning and collection of teachers' assessments.** Task distribution is reviewed once a month; the integration of the ML service with the queue is taken by the backend developer; if the schedule slips, stage 2 is cut, not the MVP.
- **The MVP is not finished in time.** The MVP core is recording upload, analysis and reports. If the schedule slips, the screens are simplified, not the pipeline; stage 2 is cut in this order: notifications and weekly summaries, then the index chart and risk, then question bank import, then the tests themselves.

10. Control and acceptance procedure

The System is accepted if the end-to-end scenario "recording upload → reports for three roles → knowledge map" passes on real recordings of individual lessons, and the NFR-5 metrics are reached on the test set. Stage 2 is accepted by the test scenarios and by the model comparison report (FR-16.5).

10.1. Test data set

- At least 20 recordings of real lessons in 2-3 subjects with the participants' consent: at least 10 in English (of which at least 5 with mixed English-Russian speech) and at least 10 in Russian. Among them are recordings with background noise and with a poor microphone.
- For each recording a reference text (selectively, at least 10 minutes) and speaker labels are prepared by hand. For 5-10 recordings the teacher uploads the material beforehand and marks for each topic whether the student answered correctly; this is the control set (FR-16.2).
- Another 3-5 recordings of Kazakh lessons - only for WER measurement (Section 3.3); they are not part of acceptance.

10.2. Types of tests

Table 13 - Types of tests

Type	What is checked	Who runs it
ML pipeline tests	WER, DER, noise suppression, teacher identification, quality of LLM analysis and the effect of recognition errors on it (NFR-5.1 - NFR-5.6)	David Khamitskiy
Fine-tuned model test	Step 2: comparison with the base model on open data (FR-16.0). Stage 2: base, fine-tuned and cloud models on the control set (NFR-5.7, NFR-5.8, FR-16.5)	Whole team
Functional testing	All MVP-priority FR requirements - by checklist	Whole team
Load and security	10 recordings in a row, report time, API response (NFR-1); access between roles and organizations, access to files by link (NFR-3)	Tairkhan Yermek, Yermakhan Khamitov
Pilot	2-3 teachers for 2-4 weeks, satisfaction survey	David Khamitskiy

10.3. End-to-end acceptance scenarios

The scenarios below are described in detail, with pass criteria, in the Verification Plan (S1-S12).

- A tutor registers an organization, adds a student and invites a parent; the parent gives consent.
- The teacher uploads material for a subject (PDF) and confirms the topics, then a 60-minute video of an English lesson; the report is ready within 30 minutes, the

utterances are labelled correctly, including those in Russian; after the speaker labels are corrected the report is recalculated; the teacher changes the status of one topic, and the index and map are updated.

- The teacher, the student and the parent see their own version of the report; a click on a timestamp opens the right moment of the recording; the student's knowledge map was updated after the lesson. The parent of another student cannot open someone else's report, either through the interface or by a direct API request.
- (Stage 2) The student receives a test on topics with an index below 80 % and takes it; the index and topic colours are updated. After the third consecutive lesson with a low index, the topic becomes "stuck". On Monday the parent and the teacher receive the weekly summary.
- (Stage 2) The system administrator switches the analysis to the fine-tuned model; a new lesson is processed by it, and the report shows the model version.

10.4. Results for the thesis

- Comparison of noise suppression models by WER; WER (English, Russian, mixed speech), DER and teacher identification accuracy; WER measurement for Kazakh.
- Comparison of 2-3 cloud LLMs by agreement with the teacher's assessment and share of valid responses; comparison of LLM analysis on verified and recognised text; results of the trial and main fine-tuning (base, fine-tuned and cloud model); pilot results.

11. Development perspectives

Outside the scope of the work; the codes are kept from Section 6.

- Kazakh language: a model fine-tuned on Kazakh speech (for example, ISSAI) after Kazakh lesson recordings are collected; then mixed Kazakh-Russian speech, reports and interface in Kazakh (FR-6.9, FR-9.8).
- Fine-tuning of the noise suppression and recognition models; regular fine-tuning of the analysis model (FR-5.6, FR-16.7).
- Uploading separate audio tracks, recording a lesson in the browser, Zoom and Google Meet, resumable upload (FR-4.8 - FR-4.10); group lessons: matching voices to students and automatic recognition by voice; manual editing of transcript text (FR-6.5, FR-6.8, FR-6.10).
- Open questions checked by the LLM, a shared question bank (FR-10.9); a question to the teacher about a report (FR-12.4, FR-13.5).
- Login with Google, importing students from CSV, programme templates (FR-1.6, FR-3.8, FR-3.9); notifications in Telegram, email and push (FR-14.2).
- Group lessons (FR-6.5); next-lesson plan from the LLM (FR-11.4); analytics for the administrator (FR-11.6); topic report (FR-9.6); pre-lesson summary (FR-11.3); organization management (FR-15.1).
- Risk prediction by an ML model, teacher effectiveness analytics, report export to PDF, monthly summary, mobile app (FR-8.8, FR-11.7, FR-9.9).

– Assessment of pronunciation, overall language level and the share of student speech in a lesson (request from an interview, Evidence of Relevance)

12. Documentation and pilot preparation

The following are created: the explanatory note of the thesis (Section 10.4), the API description in OpenAPI, a README with Docker startup, and a user guide.

References

[1] Z. Makhataeva, N. Atymtay, R. Meiramov, G. Nauryzbaikyzy, K. Sadirova, H. A. Varol. Multilingual automatic speech recognition interface for typing: Usability study and performance evaluation for Kazakh, Russian, and English // Scientific Journal of Astana IT University. 2025. Vol. 24. P. 49-63. DOI: 10.37943/24AHNP6638.

[2] S. Mussakhojayeva, K. Dauletbek, R. Yeshpanov, H. A. Varol. Multilingual speech recognition for Turkic languages // Information. 2023. Vol. 14, No. 2. Art. 74. DOI: 10.3390/info14020074.

[3] D. Khamitskiy, T. Yermek, Y. Khamitov. Speech Recognition and Large Language Models for Monitoring Knowledge Acquisition from Teacher-Student Dialogue: A Literature Review. Astana IT University, 2026.

[4] J. Macina, N. Daheim, S. P. Chowdhury, T. Sinha, M. Kapur, I. Gurevych, M. Sachan. MathDial: A Dialogue Tutoring Dataset with Rich Pedagogical Properties // Findings of EMNLP 2023. URL: https://aclanthology.org/2023.findings-emnlp.372/

[5] D. Demszky, H. Hill. The NCTE Transcripts: A Dataset of Elementary Math Classroom Transcripts // BEA 2023. P. 528-538. URL: https://aclanthology.org/2023.bea-1.44

[6] T. Dettmers, A. Pagnoni, A. Holtzman, L. Zettlemoyer. QLoRA: Efficient Finetuning of Quantized LLMs // Advances in Neural Information Processing Systems 36 (NeurIPS 2023). URL: https://arxiv.org/abs/2305.14314

[7] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen. LoRA: Low-Rank Adaptation of Large Language Models // International Conference on Learning Representations (ICLR 2022). URL: https://arxiv.org/abs/2106.09685